

Harsha Gangammanavar

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Education	Ohio State University	Columbus, OH
	Ph.D. Integrated Systems Engineering, <i>Dissertation Title:</i> Multiple Timescale Stochastic Optimization with Application to Integrating Renewable Resources in Power Systems	August 2013
	M.S. Electrical and Computer Engineering	December 2009
	Visvesvaraya Technological University	Bangalore, India
	B.E. Electronics and Communications Engineering	May 2007
Appointments	Southern Methodist University	Dallas, TX
	Operations Research and Engineering Management	
	<i>Associate Professor</i>	2023 - present
	<i>Assistant Professor</i>	2016 - 2023
	Clemson University	Clemson, SC
	Industrial Engineering	
	<i>Postdoctoral Fellow</i>	2015 - 2016
	<i>Adjunct Assistant Professor</i>	2017 - 2021
	University of Southern California	Los Angeles, CA
	Industrial and Systems Engineering	
	<i>Visiting Assistant Professor</i>	2013 - 2015
	Ohio State University	Columbus, OH
	Integrated Systems Engineering	
	<i>Graduate Research and Teaching Assistant</i>	2010 - 2013
Research Interests	<i>Methodologies:</i> Operations research; stochastic programming; large-scale optimization. <i>Applications:</i> Renewable energy integration in power systems; healthcare logistics; communication networks.	
Journal Publications	J1. D. Cole, F. Pecci, O. J. Guerra, H. Gangammanavar, J. D. Jenkins, and V. M. Zavala, Graph-Based Modeling and Decomposition of Hierarchical Optimization Problems, <i>Mathematical Programming Computation</i> , 2026; (DOI).	
	J2. N. Fadavi* and H. Gangammanavar, Active Set-based Inexact Proximal Bundle Algorithm for Stochastic Quadratic Programming, <i>Computational Optimization and Applications</i> , 93, 489–521, 2026; (DOI).	
	J3. M. Ahn, H. Gangammanavar, and D. Troxell†, Tractable Continuous Approximations for Constraint Selection via Cardinality Minimization, <i>INFORMS Journal on Optimization</i> , 7(3):195-213, 2025; (DOI).	
*Graduate student advisee		
†Undergraduate student advisee		
‡Ph.D. student coauthor/mentee		
§Postdoctoral advisee		

- J4. Sakhavand, N. and Gangammanavar, H., Subproblem sampling vs. scenario reduction: efficacy comparison for stochastic programs in power systems applications, *Energy Systems*, 16, 111-139, 2025; (DOI).
- J5. N. Sakhavand[‡], J. Rosenberger, V. Chen, and H. Gangammanavar, Design of Experiments for the Stochastic Unit Commitment with Economic Dispatch Models, vol. 12, 100089, *European Journal on Computational Optimization*, 2024; (DOI).
- J6. Z. Hoobakht[‡], H. Gangammanavar and D. Rajan, Optimal Spectral Allocation in Citizens Broadband Radio Service, *IEEE Transactions on Cognitive Communications and Networking*, vol. 10, no. 3, pp. 783-793, June 2024; (DOI).
- J7. H. Gangammanavar and M. Bansal, Stochastic Decomposition Method for Two-Stage Distributionally Robust Linear Optimization, *SIAM Journal on Optimization*, vol. 32, issue 3, pp. 1901-1930, 2022; (DOI).
- J8. D. Wood[‡], S. Çetinkaya, H. Gangammanavar, L. Weiguo, and J. Wang, On the Value of a Multistage Optimization Approach for Intensity-Modulated Radiation Therapy Planning, *Physics in Medicine and Biology*, vol. 67, no. 14, July 2022; (DOI).
- J9. S. Ariyaratne^{*}, H. Gangammanavar, and R. Sundararajan, Change Point Detection in Nonstationary Sub-Hourly Wind Time Series, *Applied Energy*, vol. 310, 118501, March 2022; (DOI).
- J10. S. Atakan, H. Gangammanavar, and S. Sen, Stochastic Hierarchical Planning for High Renewable Power Systems, *European Journal on Operational Research*, vol. 302, issue 1, pp. 381-391, October 2022; (DOI).
- J11. D. Troxell[‡], H. Ahn, and H. Gangammanavar, A Cardinality Minimization Approach to Security-Constrained Economic Dispatch, *IEEE Transactions on Power Systems*, vol. 37, no. 5, pp. 3642-3652, September 2022; (DOI).
- J12. A. Alobaidi[‡], M. Khodayar, A. Vafamehr, H. Gangammanavar, and M. Khodayar, Security Constrained Expansion Planning of Battery Energy Storage in Distribution Network with Data Centers, in *International Journal of Electrical Power and Energy Systems*, vol. 133, 107231, December 2021; (DOI).
- J13. H. Gangammanavar and S. Sen, Stochastic Dynamic Linear Program: A Distribution-free Multistage Stochastic Programming Algorithm, *SIAM Journal on Optimization*, vol. 31, issue 3, pp. 2111-2140, 2021; (DOI).
- J14. S. Yin[‡], J. Wang, and H. Gangammanavar, Stochastic Market Operation for Coordinated Transmission and Distribution Systems, in *IEEE Transactions on Sustainable Energy*, vol. 12, no. 4, pp. 1996-2007, October 2021; (DOI).
- J15. H. Gangammanavar, Y. Liu, and S. Sen, Stochastic decomposition for two-stage stochastic linear programs with random cost coefficients, *INFORMS Journal on Computing*, vol. 33, no. 1, pp 51–71, January 2021; (DOI).
- J16. S. Wang[‡], S. J. Mason, and H. Gangammanavar, Stochastic optimization for flow-shop scheduling with on-site renewable energy generation using a case in the United States, *Computers and Industrial Engineering*, vol. 149, 106812, November 2020; (DOI).
- J17. S. Wang[‡], H. Gangammanavar, S. Ekşioğlu, and S. J. Mason, Statistical estimation of operating reserve requirements using rolling horizon stochastic optimization, *Annals of Operations Research*, vol. 292, issue 1, pp. 371–397, 2020; (DOI).
- J18. Z. Azadi[‡], H. Gangammanavar, and S. Ekşioğlu, Developing childhood vaccine administration and inventory replenishment policies that minimize open vial wastage. *Annals of Operations Research*, vol. 292, issue 1, pp. 215–247, 2020; (DOI).
- J19. S. Wang[‡], H. Gangammanavar, S. D. Ekşioğlu, and S. J. Mason, Stochastic Optimization for Energy Management in Power Systems With Multiple Microgrids, in *IEEE Transactions on Smart Grid*, vol. 10, no. 1, pp. 1068-1079, Jan. 2019; (DOI).

- J20. H. Gangammanavar and S. Sen, Two-scale Stochastic Optimization for Controlling Distributed Storage Devices, in *IEEE Transactions on Smart Grid*, vol. 9, no. 4, pp. 2691-2702, July 2018; (DOI).
- J21. H. Gangammanavar, S. Sen, and V. M. Zavala, Stochastic Optimization of Sub-Hourly Economic Dispatch With Wind Energy, in *IEEE Transactions on Power Systems*, vol. 31, no. 2, pp. 949-959, March 2016; (DOI).
- J22. R. Li, H. Gangammanavar, and A. Eryilmaz, Optimal Dynamic Coding and Rate-Control for Serving Deadline-Constrained Traffic over Time-Varying Channels, in *IEEE Transactions on Information Theory*, vol. 58, no. 10, pp. 6556-6571, 2012; (DOI).
- Book Chapter** B1. H. Gangammanavar, Sampling-based Decomposition Algorithms for Multistage Stochastic Programming, *Encyclopedia of Optimization*, 2022.
- Conference Proceedings** C1. K. Baker and H. Gangammanavar, Locational Marginal Prices Obey DC Circuit Laws, accepted for publication in the *Proceedings of the PowerUp Conference*, Boulder, CO, September 2026.
- C2. J. Forner*, M. Ahn, and H. Gangammanavar, Decision-Aware Predictions for Right-Hand Side Parameters in Linear Programs, accepted for publication in the *Proceedings of the INFORMS Optimization Society Conference*, Atlanta GA, March 2026 (DOI).
- C3. Z. Azadi†, H. Gangammanavar, and S. D. Ekşioğlu, Stochastic Optimization for Vaccine Vial Replenishment, in *Proceedings of the 2016 Industrial and Systems Engineering Research Conference (ISERC)*, Anaheim, CA.
- C4. H. Gangammanavar and A. Eryilmaz, Dynamic Coding and Rate-Control for Serving Deadline-Constrained Traffic over Fading Channels, in *Proceedings of IEEE International Symposium on Information Theory (ISIT)* Austin TX, pp. 1788–1792, 13-18 June 2010.
- Under Review** R1. Z. Wang*, S. Ariyaratne*, H. Gangammanavar, and J. Wang, Multiagent Optimization for Coordinated Operations of Transmission and Distribution System, 2026 (first review). .
- R2. J. Zhou, H. Gangammanavar, and Y. Xiang, Centralized Training for Partially Decentralized Execution in Multi-Agent Reinforcement Learning, 2026 (first revision).
- R3. C. Sharma§ and H. Gangammanavar, Data-driven Policies for Two-stage Stochastic Linear Programming, 2026 (first review).
- Working Papers** W1. C. Sharma§ and H. Gangammanavar, Multiagent Stochastic Programming.
- W2. S. Ariyaratne* and H. Gangammanavar, New Formulations and Pricing Mechanisms for Stochastic Electricity Market Clearing Problem, 2026.
- W3. K. Baker and H. Gangammanavar, Relationship of Locational Marginal Prices to Network Properties and its Implications.
- Unpublished** U1. S. Tabrizian*, H. Gangammanavar, and H. Üster, An Adaptive Cluster Sampling-based Solution Method for Two-stage Stochastic Linear Programs, 2020.

Grants (External)

- G1. *XTRIPODS: Integrated Decision-Making in Data Science: New Models, Tractable Reformulations, and Validation through Undergraduate Projects*; Role Co-PI (with M. Ahn); National Science Foundation; #2423246; Total award amount: \$199,698; October 2024 - September 2026.
- G2. *Integrated Framework for Cooperative 3D Printing: Uncertainty Quantification, Decision Models, and Algorithms*; Role: Co-PI (with Y. Xiang, PI and W. Zhou, Co. PI); National Science Foundation; #2329739; Total award amount: \$505,789; January 2024 - December 2026.
- G3. *New Abstractions and Randomized Algorithms for Multiscale Stochastic Optimization*; Role: Lead PI; Department of Energy - Office of Science; #DE-SC0023361; Total award amount: \$2,040,256; October 2022 - September 2025.
- G4. *Stochastic Programming Decomposition Models and Algorithms for Discrete-event Dynamic Systems*; Role: Sole PI; Office of Naval Research #N00014-22-1-2603; \$387,826; September 2022-August 2025.
- G5. *Statistical Optimality, Algorithms and Resilience in Time-Staged Stochastic Systems*; Role: Co-PI (with S. Sen, PI); Air Force Office of Scientific Research #FA9550-15-1-0267; \$450,000; August 2015 - December 2018.

Grants (Internal)

- G6. *Data-driven Multistage Decision Policies: Integration of Optimization and Statistical Learning*; Role: PI; SMU Provost's Science and Engineering Postdoctoral Bridge Grant; Award amount: \$70,000; January 2024 - January 2025.
- G7. *Data Assimilation for Radiation Therapy Planning via Optimization: Adaptive Deterministic Models*; Role: PI (with S. Çetinkaya); SMU Lyle School Research Seed Funding; \$30,500.00; March - December 2020.
- G8. *Multi-temporal Flexibility Services in Transactive Energy Architecture*; Role: Co-PI (with M. Khodayar, PI); SMU Lyle School Research Seed Funding; \$23,760.00; March - December 2018.
- G9. *A Data-Driven Support System for Coordinated Operation of Electricity and Natural Gas Infrastructure*; Role: PI (with M. Khodayar); SMU Lyle School Research Seed Funding; \$25,080.00; March - December 2017.

Other Research Engagements

- Member, Electrification Council, NSF Engines Development Award: Advancing logistics technologies, May 2023 - April 2025.
- Educational Experience Agreement with Radiation Oncology MAIA Laboratory, University of Texas Southwestern (UTSW Contract ID # 2020—7031), January 2020—January 2021. Collaborative research with S. Çetinkaya and J. Wang (UTSW).

Honors

- Honorable mention at Undergraduate Operations Research Prize (for UG student D. Troxell; joint work with M. Ahn), INFORMS Annual Meeting, Anaheim, Oct. 2021;
- Fellow of the Dedman College Interdisciplinary Institute 2017-18, Southern Methodist University;
- Honorable mention at Minority Issues Forum poster competition (for Z. Azadi*; joint work with S. Eksioğlu), INFORMS Annual Meeting, Nashville, Nov. 2016;
- Postdoctoral Fellowship, Clemson University, 2015-16;
- Travel grant recipient, PhD Winter School on Managing Uncertainty in Energy Infrastructure Investment, Oppdal, Norway, 2011;
- Travel grant recipient, Illinois Wireless Summer School, University of Illinois at Urbana-Champaign 2009.

**Courses
Taught**

At Southern Methodist University.

- OREM 2360 Engineering Economy (UG): Spring 2026 (15);
- OREM 3360 Operations Research (UG): Spring 2017 (26), 2018 (20), 2019 (24), 2022 (19), 2023 (7), 2024 (20); Fall 2021 (25), 2022 (13);
- OREM 5364 Advanced Operations Research (UG): Fall 2022 (10), 2023 (5), 2025 (15); Spring 2024 (8);
- OREM 8360 Operations Research Models (G): Fall 2016 (26), 2017 (13), 2018 (15), 2020 (4); Spring 2020 (10), 2021 (2);
- OREM 8371 Linear Programming (G): Fall 2018 (6), 2019 (9); Spring 2022 (10);
- OREM 8384 Stochastic Programming (G): Spring 2018 (13), 2020 (6); Fall 2024 (8), 2025 (5);

At University of Southern California

- ISE 310 Facilities and Logistics (UG): Spring 2015;
- ISE 330 Introduction to Operations Research: Deterministic Models (UG): Spring 2015, 2014; Fall 2014, 2013;
- ISE 499 Special Topics: Integrative Systems Engineering (UG): Spring 2015, 2014;
- ISE 536 Linear Programming and Extensions (G): Fall 2014.

**Postdoctoral
Supervision**

- Chhavi Sharma;
March 2024 - August 2025;

**Ph.D.
Supervision**

- Ishara Aluthwala, Ph.D. Student in Operations Research;
2023 - present.
- Jackson Forner, Ph.D. Student in Operations Research;
2023 - present, SMU (co-advised with Prof. Miju Ahn)
- Zhixue Wang, Ph.D. Student in Operations Research;
2025 - present.
- Zhiyuan Zhang, Ph.D. Student in Operations Research;
2024 - present.
- Niloofar Fadavi, PhD in Operations Research, SMU;
Recipient of Center for Research Computing Fellowship;
Graduated: December 2024
Dissertation title: Inexact methods for Large-scale Stochastic Programming;
First position: Senior Operations Research Scientist, Optym.
- Sakitha Ariyaratne, Ph.D. in Operations Research, SMU;
Graduated: December 2022
Dissertation title: Study of Stochastic Market Clearing Problem in Power Systems With High Renewable Integration;
First position: Data Scientist, BHG Financial.
- Siavash Tabrizian. Ph.D. in Operations Research, SMU;
Recipient of University Ph.D. Fellowship;
Graduated: December 2021 (co-advised with Prof. Halit Uster);
Dissertation title: Sampling-based Algorithms for Two-stage Stochastic Programs and Applications;
First position: Data Scientist at USX Variant, Atlanta, GA.

Masters Supervision

- Nahal Sakhavand, M.S. in Operations Research
Graduated: Summer 2018;
First position: Ph.D. student in ISME, University of Texas at Arlington.

Undergraduate Mentoring

- Ethan Yang, B.S. in Management Science, Summer & Fall 2024;
- Brody Thalmann, B.S. in Management Science, Spring 2024;
- Emily Davidson, B.S. in Management Science, Spring & Summer 2024;
- David Troxell, B.S. in Management Science, Fall 2020 & Spring 2021;
First position: M.S. student in Data Science, Stanford University.

Graduate Committee Service

Program	Students
Ph.D. (OREM/EMIS)	Ongoing: Mohammad Khalafi; Completed: Toby Huskinson (2025); Chengyu Ke (2023); Hedieh Ashrafi (2021); Justin B. Brown (2021); Naderehsadat Mansouri (2019); Amin Ziaefar (2019).
Ph.D. (ECE)	Ongoing: Xinyun Lu; Tao Wu; Completed: Abduraheem Alobaidi (2022); Yazeed Alkhrijah (2022); Shengfei Yin (2021); Mahdi Khodayar (2020); Xinan Wang (2020); Ying Zhang (2020).
Praxis (OREM/EMIS)	Emily McIntosh (2022); Mohammed Abdul Qaudeer (2020); Peng Yang (2019).
Ph.D. (UTA IMSE)	Nahal Sakhavand (2021).
Ph.D. (Clemson IE)	Shasha Wang (2020); Site Wang (2018).

Professional Society Service

- Secretary, Committee on Stochastic Programming 2025 – present;
- Board member, INFORMS Computing Society, 2023-2025.
- *Organization Committee Member:*
 - Organizing committee member and co-chair of contributed sessions tracks at INFORMS Annual Meeting 2023;
 - Program committee member (Energy Systems Track) at IISE Annual Conference and Expo 2022;
 - NSF Operations Engineering Workshop, SMU, March 2019.
- *Conference Session Chair:*
 - INFORMS Annual Meetings 2025 (Atlanta), 2023(Phoenix), 2022 (Indianapolis), 2021 (Anaheim), 2019 (Seattle), 2018 (Phoenix), 2014 (San Francisco).
- *Referee:* INFORMS Operations Research, Journal on Computing, and Journal on Optimization; SIAM Journal on Optimization; Mathematical Programming; Computational Optimization and Applications; European Journal on Computational; IISE Transactions; Optimization Letters; Energy Systems; Omega: International Journal of Management Science; Stochastic Models; Naval Research Logistics; SMU Journal of Undergraduate Research; IEEE Transaction on Power Systems, Transactions on Smart Grid, and Transactions on Sustainable Energy; Electric Power Systems Research; IET Generation, Transmission and Distribution.
- *Panelist:* National Science Foundation, 2023, 2017; Office of Naval Research, 2024, 2023, 2022; Department of Energy - Office of Science, 2024, 2023.

- *Award Committee Member:* IISE Energy Systems Best Paper Award 2022; George Nicholson Student Paper Competition, INFORMS 2020 and 2021; INFORMS-ENRE Student Paper Competition, INFORMS, 2018.
- *Faculty Advisor:* SMU INFORMS Student Chapter, 2018-2022.
- *Vice-President:* Ohio State University INFORMS Student Chapter, 2011-2012.

**Professional
Society
Membership**

- Institute for Operations Research and Management Science (INFORMS): Optimization Society; Computing Society; and Energy, Natural Resources, and Environment Society.
- Society of Industrial and Applied Mathematics (SIAM).
- Mathematical Optimization Society (MOS).

**Administrative
Service**

- Member, Mathematics Faculty Search Committee, Fall 2026.
- Member and Chair, OREM Faculty Search Committee, Spring 2026.
- OREM/EMIS department representative on Lyle Curriculum Committee (formerly, Academic Affairs Committee), 2021 – 2022; 2025 – present;
- Member, Lyle Faculty Search Committee, 2023 – 2024 (Endowed position in the Energy cluster); 2025 – present (Endowed position on Advanced Manufacturing);
- Lyle School representative, SMU Faculty Senate; member, Economic Status committee, 2023 – present;
- Member, Faculty advisory group, Data Science Institute, 2022 – present;
- Member and chair, OREM Promotion and Tenure committee, Fall 2024;
- Member, OR Program Recruiting and Marketing Committee, 2023 – 2024;
- Member and chair, OREM Graduate Committee, 2022 – 2024;
- EMIS/OREM Department Seminar Organizer, 2020 – 2023;
- Member and chair, EMIS Course Coordination Committee on OR Methods, 2019 – 2022;
- Member, EMIS Accreditation and Academic Programs, 2020 – 2021;
- Member, EMIS Faculty Search Committee, 2018-19, 2019-20, 2021-22;
- Instructor, Summer Bit Blast, workshop organized by the Center for Research Computing (SMU), July 2022;
- Judge, Dallas Regional Science and Engineering Fair, 2020 and 2021;
- Judge, Lyle Research Day, 2016, 2017, 2018, and 2019.

**Invited
Seminars**

- Physics and Engineering Department, Austin College, Sherman, TX, 2025.
- Systems and Industrial Engineering Seminar, University of Arizona, Tucson, November, 2023.
- Decision Sciences Area Seminar, Indian Institute of Management, Bangalore, December 2021.
- Industrial Engineering, University of Houston, March 2021.
- Industrial, Manufacturing, and Systems Engineering, University of Texas at Arlington, December 2018.
- Center for Applicable Mathematics, Tata Institute of Fundamental Research, Bangalore, July 2018.
- Department of Mechanical Engineering, University of Texas at Dallas, June 2018.

- Dedman College Interdisciplinary Institute (DCII), Operations Research and Statistics Cluster towards Integrative Analytics, SMU, February 2017.
- Department of Engineering Management, Information, and Systems, Southern Methodist University, February 2016.
- Industrial Engineering Technical Innovation Seminar Series, Clemson University, November 2015.
- Ming Hsieh Department of Electrical Engineering, University of Southern California, October 2014.
- Daniel J Epstein Department of Industrial and Systems Engineering, University of Southern California, October 2014.

Conference Presentations

- *New Formulations and Pricing Mechanisms for Stochastic Electricity Market Clearing Problem* with S. Ariyaratane,
 - INFORMS Annual Meeting 2023, Phoenix, AZ, October 2023.
- *Stochastic dynamic linear programming: A sequential sampling algorithm* with S. Sen,
 - Invited Semi-Plenary: International Conference on Stochastic Programming, Davis, CA, June 2023.
 - INFORMS Optimization Society Conference, Greenville, SC, March 2020.
 - 21st Conference of the International Federation of Operational Research Societies, Quebec City, Canada, July 2017,
 - SIAM Conference on Optimization, Vancouver, Canada, May 2017.
- *Stochastic Branch-and-Cut Algorithm from a Sequential Sampling Perspective*
 - SIAM Conference on Optimization, Seattle, WA, May 2023,
- *Sequential sampling-based solution algorithms for distributionally robust optimization*,
 - International Conference on Continuous Optimization, Lehigh, PA, July 2022.
 - International Conference on Stochastic Programming, Trondheim, July 2019.
- *A sampling-based branch-and-cut algorithm for two-stage stochastic mixed-integer programming*,
 - INFORMS Annual Meeting, Anaheim, 2021.
- *Stochastic decomposition for two-stage stochastic linear programs with random cost coefficients*,
 - INFORMS Annual Meeting, Phoenix, Nov. 2018.
- *Stochastic programming framework for coordinated operation of power systems with multiple microgrids*,
 - International Symposium on Mathematical Programming, Bordeaux, July 2018.
 - INFORMS Optimization Society Conference, Denver, March 2018.
- *Sequential sampling-based optimization for power systems application*, INFORMS Annual Meeting, Nashville, Nov. 2016.
- *Convergence proofs of SDDP and multistage stochastic decomposition* with S. Sen, International Conference on Stochastic Programming, Buzios, Brazil, June 2016.
- *Multiple timescale stochastic optimization for integrating renewable resources* with S. Sen:
 - INFORMS Annual Meeting, San Francisco, Nov. 2014

- Workshop on Optimization Under Uncertainty: Energy, Transportation and Natural Resources, University of California-Davis, Nov. 2014
- Smartgrid Challenges, University of Arizona, Tucson, Mar. 2013.
- *Stochastic optimization of sub-hourly economic dispatch with wind generation*
 - INFORMS Annual Meeting, San Francisco, Nov. 2014
 - INFORMS Annual Meeting, Minneapolis, Oct. 2013.
- *Dynamic coding and rate-control for serving deadline-constrained traffic over fading channels*, with A. Eryilmaz, IEEE International Symposium on Information Theory (ISIT), Austin, Jun. 2010.

**Workshops
Attended**

- “Deep Learning”, 25th Annual Teaching Effectiveness Symposium, Center for Teaching Excellence, Southern Methodist University, August 2017.
- New Faculty Colloquium, INFORMS Annual Meeting 2016, Nashville, October 2016.
- “A Conversation between Artificial Intelligence, Operations Research and Control Theory on Stochastic Optimization”, NSF Workshop at Rutgers University, 2012.
- “Managing Uncertainty in Energy Infrastructure Investments”, Ph.D. Winter School, Oppdal, Norway, 2011 (recipient of workshop travel grant).
- Ph.D. Workshop at 12th International Conference on Stochastic Programming, Halifax, NS, Canada, 2010.
- Illinois Wireless Summer School, University of Illinois, Urbana-Champaign, IL, 2010 (recipient of summer school travel grant).